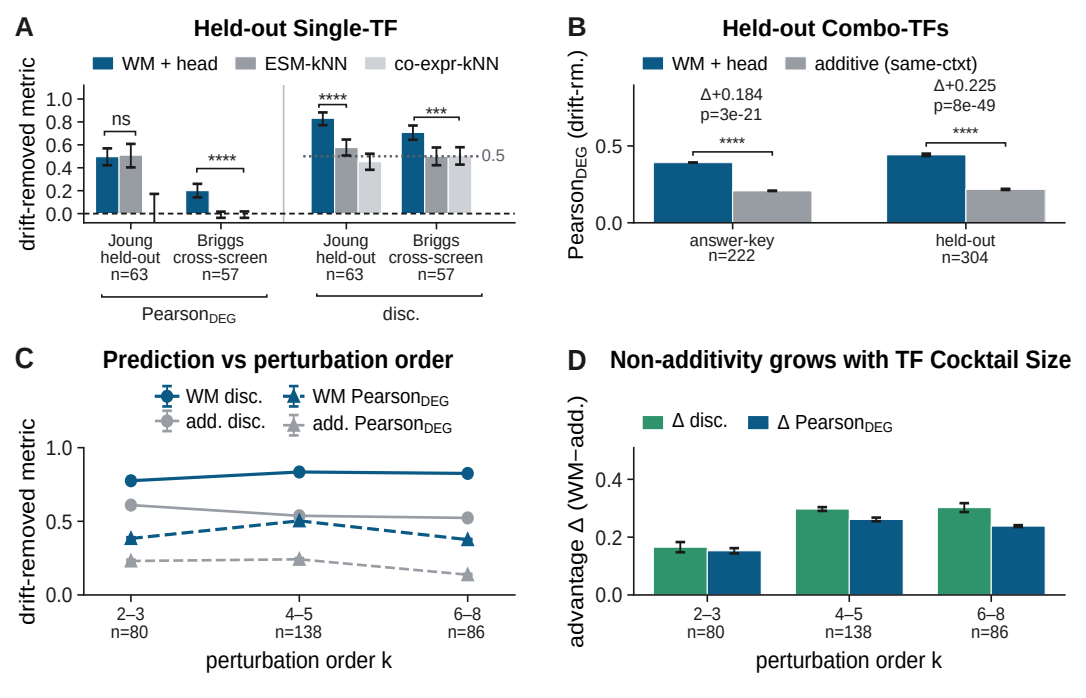

A
Held-out Single-TF
WM + head
ESM-kNN
co-expr-kNN
drift-removed metric
1.0
0.8
0.6
0.4
0.2
0.0
ns
****
****
***
0.5
Joung
held-out
n=63
Briggs
cross-screen
n=57
Joung
held-out
n=63
Briggs
cross-screen
n=57
Pearson$_{DEG}$
disc.
B
Held-out Combo-TFs
WM + head
additive (same-ctxt)
Pearson$_{DEG}$ (drift-rm.)
Δ+0.184
p=3e-21
****
Δ+0.225
p=8e-49
****
0.5
0.0
answer-key
n=222
held-out
n=304
C
Prediction vs perturbation order
WM disc.
add. disc.
WM Pearson$_{DEG}$
add. Pearson$_{DEG}$
drift-removed metric
1.0
0.5
0.0
2–3
n=80
4–5
n=138
6–8
n=86
perturbation order k
D
Non-additivity grows with TF Cocktail Size
Δ disc.
Δ Pearson$_{DEG}$
advantage Δ (WM−add.)
0.4
0.2
0.0
2–3
n=80
4–5
n=138
6–8
n=86
perturbation order k